

1. What are the assumptions of linear regression regarding residuals?

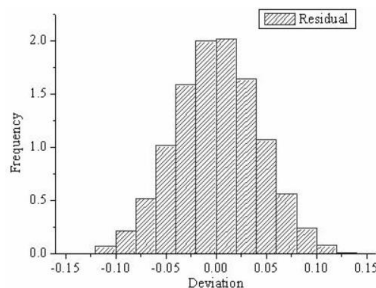
By definition Residuals is the difference between the predicted variable through the constructed model and actual target variable. Which is important measure for developing model to optimize the model development to focus in residuals. Model can be created to optimize the parameters to model for completely reducing residuals, however this will not suitable for unseen data. Important consideration is to create model with more generalization hence even residuals are little higher side the model performance on the training data set is will be very low.

By considering model to generalized following assumptions should be considered while analyzing residuals.

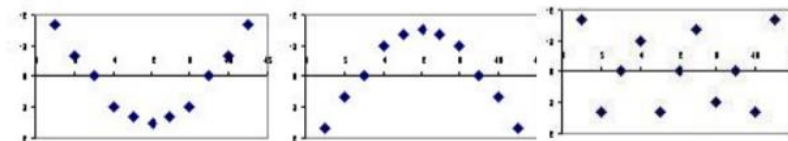
Definition of Error Terms: Difference between actual target value – predicted values through model.

Fundamental assumptions of Linear Regressions

- There is a linear relation between dependent and independent variable
- Error terms are normally distributed: which means there are no bias in prediction model fit variance of error terms are equally spread across data point. This can be validated with the plotting histogram, for example below represents normal distributed of error terms.



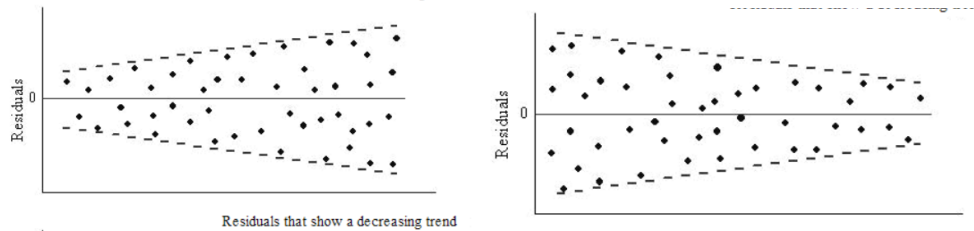
- Error Terms are independent for each other: represent that in error there is no specific series and will not depending upon either side of neighboring data points. This can be verified by plotting scatter plot of error terms in time horizon.



For example on the above figure -1 and figure -2 represent specific trend which may be influence of previous error terms. However, in the third picture it clearly evidencing that error terms are no trend line.

- Error terms have constant variance:

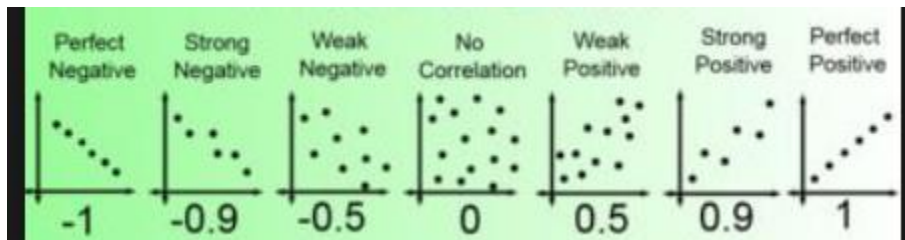
This represents that variance is constant across the target variables. In contrast, if error terms either increase or decrease based on the target variable, this represents



For example, in the above two graphs, residuals are continuously increasing with the decrease, which represents that the model is not properly fit. Usually, it is preferred to have “homoscedasticity” where residuals are required to have a constant variance, which represents no bias to the independent variable.

2. What is the coefficient of correlation and the coefficient of determination?

Coefficient of correlation represents a statistical measure of correlation between dependent and independent variables. Statistically, the coefficient will be between -1 and 1. If the coefficient value is -ve, it means that when the independent variable increases, the dependent variable decreases. Vice versa, when the independent variable decreases, the dependent variable will increase. In other words, +ve values represent a +ve relationship between dependent and independent variables. If the correlation is equal to zero, it indicates no relationship between dependent variables. When the correlation reaches towards 1, it indicates a stronger relationship.



For example, in the above graph, it indicates the strong -ve correlation to words no correlation and strong +ve correlation.

It is also represented with the variable “r” and a real number between -1 and 1.

Correlation Determination: represented with the R^2 . It usually is between 0 and 1. Correlation determination represents how much portion of data points are explained by the model. Determination is high (close to 1) indicating the model completely explains the target variable and when R Square is near to zero, it indicates no explanation by the model.

R Square is calculated through the following: $= 1 - (RSS / TSS)$

RSS Residual Sum of Square: difference between the actual y target variable and predicted value and Square and sum for all the elements.

TSS Total Sum of Square it is calculated based on the difference between mean of target variable and predicted variable.

Problem with the R Square is increasing continuously when the new number of independent variables are adding to the model. to balance this we will monitor the adjusted R Square to apply the degree of freedom by using number of variables and sample size.

3. Explain the Anscombe's quartet in detail.

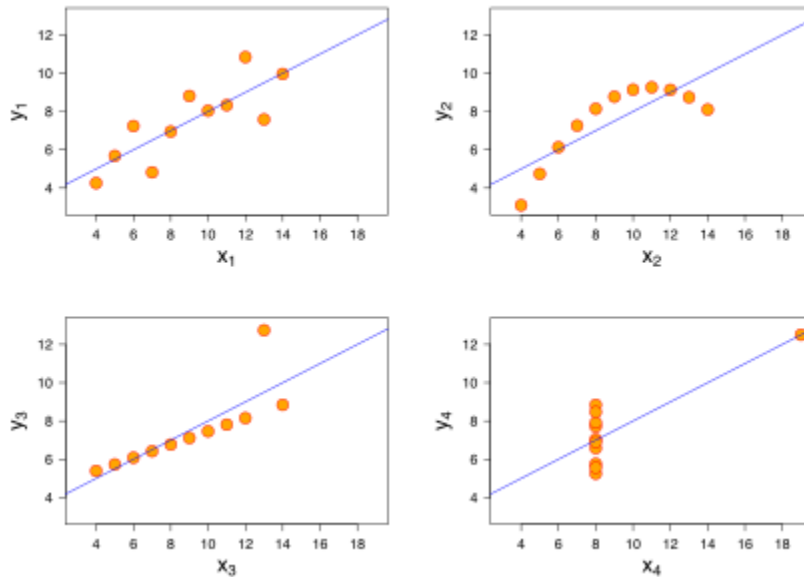
Anscombe's Quartet was developed to explain the statistical relationship of regression and best fit line. This data sets were proposed by the statistician Francis Anscombe. It contain the four data set's each contain the different relationship between' s the dependent and independent variable.

The important aspect of these data sets are they are all share the same quantities of summary statistics with the same mean of $x = 9$, standard variance of $x = 11$ and similarly mean of $y = 7.50$ and Sample variance of $y = 4.125$.

When evaluate the equation of correlation R between these 4 data sets all the results are same as shown in the below figure.

	I		II		III		IV	
	x	y	x	y	x	y	x	y
	10	8.04	10	9.14	10	7.46	8	6.58
	8	6.95	8	8.14	8	6.77	8	5.76
	13	7.58	13	8.74	13	12.74	8	7.71
	9	8.81	9	8.77	9	7.11	8	8.84
	11	8.33	11	9.26	11	7.81	8	8.47
	14	9.96	14	8.1	14	8.84	8	7.04
	6	7.24	6	6.13	6	6.08	8	5.25
	4	4.26	4	3.1	4	5.39	19	12.5
	12	10.84	12	9.13	12	8.15	8	5.56
	7	4.82	7	7.26	7	6.42	8	7.91
	5	5.68	5	4.74	5	5.73	8	6.89
Mean	9	7.500909	9	7.500909	9	7.5	9	7.500909
Std Variance	9.166667	3.439391	9.166667	3.439691	9.166667	3.435517	9.166667	3.436041
standard deviation	3.02765	1.854559	3.02765	1.85464	3.02765	1.853515	3.02765	1.853656
Correlation	0.816421		0.816237		0.816287		0.816521	

However relationship between X and Y is not the same for all these four models. All 4 models represents the different relationship even though summary statistics are same doesn't mean relationship can be same.



If you noticed that Picture -1 represents simple linear regression and corresponding data points are across to the fitted line.

In the second graph, error terms are not distributed normally there are some kind of polynomial trend between x and y . however linear model will fail to fit the line. In this case Pearson correlation coefficient indicating of 0.81 may not represent true picture in this case observing coefficient determination (R Square) will better explain the model performance.

In the 3^d picture, it is evidencing clear linear relation between X and Y . however since because of outlier In the dataset the slop of the line litter lean towards upside in the end. If this outlier is not there in the data set prediction model will be perfect fit.

In the 4th picture , fact is there is no correlation between X and Y , where X is constant and y in varying. However Pearson correlation failed to detect this variation and indicated both variables are highly correlated.

In summary, we should not take decision based one variable blindly based on the correlation R , to be cation need to review regression of determination as well residual variances to ensure model developed is performing better for unseen data.

4. What is Pearson's R ?

Pearson's r , also called as Pearson correlation coefficient, which represent the bivariate correlations between two numeric random variables. Usually the value of r between -1 and 1. Anything near to 1 indicating strong correlation between two variables and the value near to zero indicating the either no correlation are marginal correlation.

When r value +ve indicating that, if one variable increase dependent variable also increase sturdily. Same time if variable is decrease dependent variable will also decrease.

If incase r value -ve, that represent -ve correlation between dependent and independent variables. That means when the independent variable increase dependent variable go in opposite direction that means decreases. Visa versa when independent variable decreases dependent variable increases.

In python, we can use "corr" function to measure the Pearson's coefficient between two series of random numbers. Usually anything over

	Positive	Negative
Strong	1 to 0.8	-0.8 to 1
Moderate	0.8 to 0.5	-0.5 to 0.8
Weak	0.5 to 0.3	- 0.5 to -0.3
No Correlation	0.3 to 0	0 to -0.3

The above threshold's can be considered in understanding correlations between two variables.

5. What is scaling? Why is scaling performed? What is the difference between normalized scaling and standardized scaling?

Feature scaling is the processing normalizing the data points between specific ranges without disturbing the shape of the data.

Usually

6. You might have observed that sometimes the value of VIF is infinite. Why does this happen?